```
#import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
from sklearn.metrics import mean_squared_error
from sklearn.metrics import mean_absolute_error
```

```
#load dataset
df=pd.read_csv("advertising.csv")

print("Dataset Loaded Successfully")
```

```
Dataset Loaded Successfully
```

```
df.shape
```

```
(200, 4)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   TV         200 non-null    float64
 1   Radio      200 non-null    float64
 2   Newspaper  200 non-null    float64
 3   Sales      200 non-null    float64
dtypes: float64(4)
memory usage: 6.4 KB
```

```
df.head()
```

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9

```
df.describe()
```

	TV	Radio	Newspaper	Sales
count	200.000000	200.000000	200.000000	200.000000
mean	147.042500	23.264000	30.554000	15.130500
std	85.854236	14.846809	21.778621	5.283892
min	0.700000	0.000000	0.300000	1.600000
25%	74.375000	9.975000	12.750000	11.000000
50%	149.750000	22.900000	25.750000	16.000000
75%	218.825000	36.525000	45.100000	19.050000
max	296.400000	49.600000	114.000000	27.000000

```
print(data.corr())
```

```
                TV     Radio  Newspaper     Sales
TV        1.000000  0.054809   0.056648  0.901208
Radio     0.054809  1.000000   0.354104  0.349631
```

```
Newspaper  0.056648  0.354104   1.000000  0.157960
Sales      0.901208  0.349631   0.157960  1.000000
```

```
data.isnull().sum()
```

	0
TV	0
Radio	0
Newspaper	0
Sales	0

dtype: int64

```
data.duplicated().sum()
```

```
np.int64(0)
```

```
X = data[['TV']]
Y = data['Sales']
```

```
# Split Dataset
X_train, X_test, Y_train, Y_test = train_test_split(X,Y,test_size=0.2,random_state=42)
```

```
# Create Model
model = LinearRegression()
```

```
# Train Model
model.fit(X_train, Y_train)

print("Model Trained")
```

```
Model Trained
```

```
# Predictions
Y_train_pred = model.predict(X_train)
Y_test_pred = model.predict(X_test)

print("Predictions Generated")
```

```
Predictions Generated
```

```
#for training model
mae = mean_absolute_error(Y_train, Y_train_pred)
mse = mean_squared_error(Y_train, Y_train_pred)
rmse = np.sqrt(mse)
r2 = r2_score(Y_train, Y_train_pred)

print("Training Results")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
Training Results
MAE : 1.8005092256620792
MSE : 4.998442356450173
RMSE: 2.235719650683013
R2 Score: 0.8134866044709264
```

```
#for testing model
mae = mean_absolute_error(Y_test, Y_test_pred)
mse = mean_squared_error(Y_test, Y_test_pred)
rmse = np.sqrt(mse)
r2 = r2_score(Y_test, Y_test_pred)

print("Testing Results")
print("MAE :", mae)
print("MSE :", mse)
```

```
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
Testing Results
MAE : 1.9502948931650088
MSE : 6.101072906773963
RMSE: 2.470035001123256
R2 Score: 0.802561303423698
```

```
print("Slope :", model.coef_[0])
```

```
Slope : 0.0554829439314632
```

```
print("Intercept :", model.intercept_)
```

```
Intercept : 7.007108428241848
```

```
# Plot actual training data points and the fitted linear regression line
plt.figure(figsize=(8,5))

plt.scatter(X_train, Y_train, color='blue', label='Actual Data')
plt.plot(X_train, Y_train_pred, color='red', linewidth=2, label='Fitted Line')

plt.xlabel('TV Advertising Budget')
plt.ylabel('Sales')
plt.title('Actual vs. Fitted Linear Regression Line')
plt.legend()

plt.show()
```